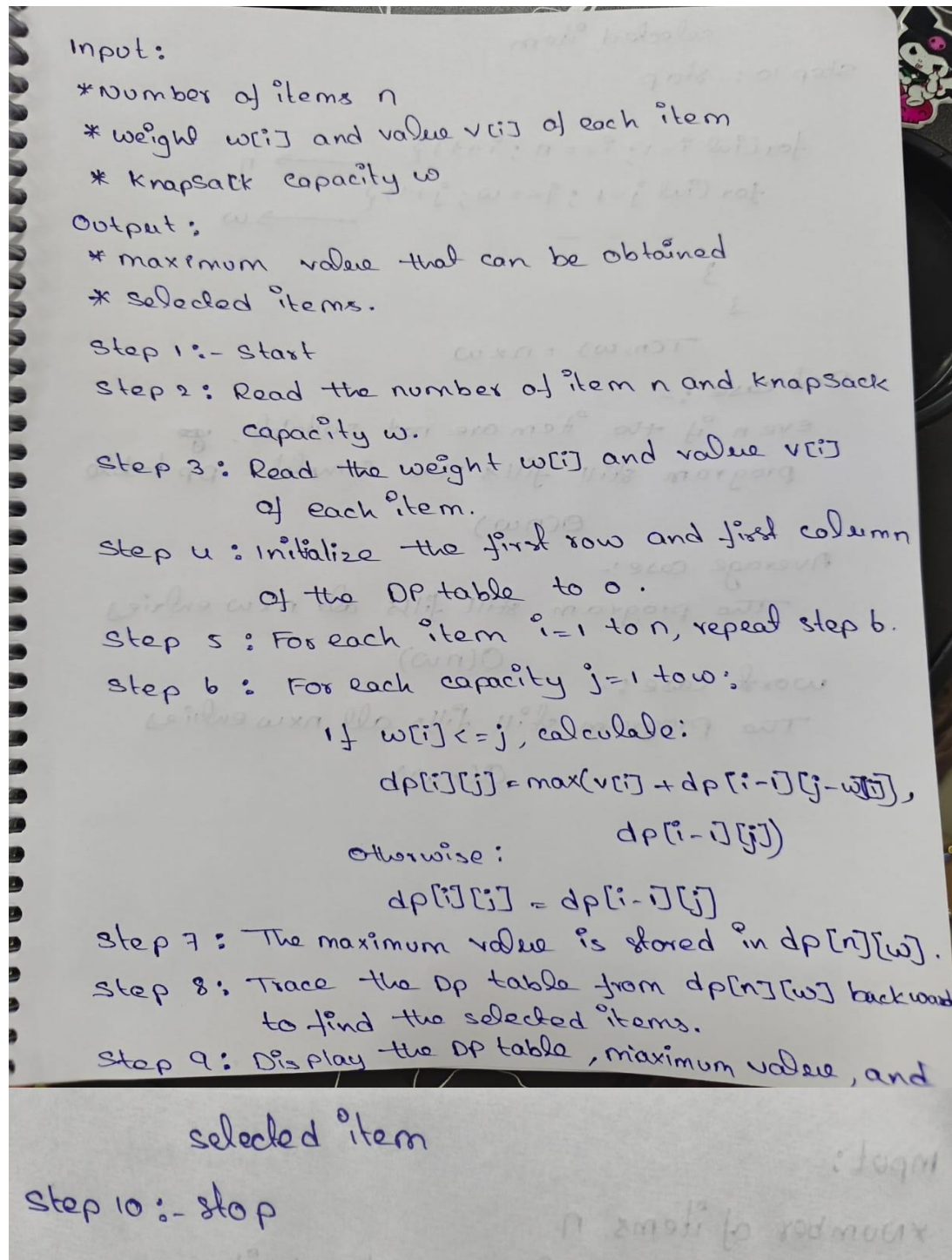


Practical- 5

Aim: To Implementation of a knapsack problem using dynamic programming.

Knapsack

Algorithm:



Input:

- * Number of items n
- * weight $w[i]$ and value $v[i]$ of each item
- * Knapsack capacity w

Output:

- * maximum value that can be obtained
- * selected items.

Step 1:- Start

Step 2: Read the number of item n and knapsack capacity w .

Step 3: Read the weight $w[i]$ and value $v[i]$ of each item.

Step 4: Initialize the first row and first column of the DP table to 0.

Step 5: For each item $i=1$ to n , repeat step 6.

Step 6: For each capacity $j=1$ to w :

- if $w[i] \leq j$, calculate:
$$dp[i][j] = \max(v[i] + dp[i-1][j-w[i]], dp[i-1][j])$$
- otherwise:
$$dp[i][j] = dp[i-1][j]$$

Step 7: The maximum value is stored in $dp[n][w]$.

Step 8: Trace the DP table from $dp[n][w]$ backward to find the selected items.

Step 9: Display the DP table, maximum value, and selected item

Step 10:- stop

Program:

```
#include<iostream>

using namespace std;

class Knapsack {
private:
    int w[10];
    int v[10];
    int n;
    int W;
    int dp[10][100];
public:
    void getInput() {
        cout << "Enter number of items  : ";
        cin >> n;
        cout << "Enter knapsack capacity : ";
        cin >> W;
        for(int i = 1; i <= n; i++) {
            cout << "Item " << i << " weight and value : ";
            cin >> w[i] >> v[i];
        }
    }
    void solve() {
        for(int i = 0; i <= n; i++) dp[i][0] = 0;
        for(int j = 0; j <= W; j++) dp[0][j] = 0;
        for(int i = 1; i <= n; i++) {
            for(int j = 1; j <= W; j++) {
                if(w[i] <= j) {
                    dp[i][j] = max(
                        v[i] + dp[i-1][j-w[i]],
                        dp[i-1][j]
                    );
                } else {
                    dp[i][j] = dp[i-1][j];
                }
            }
        }
    }
};
```

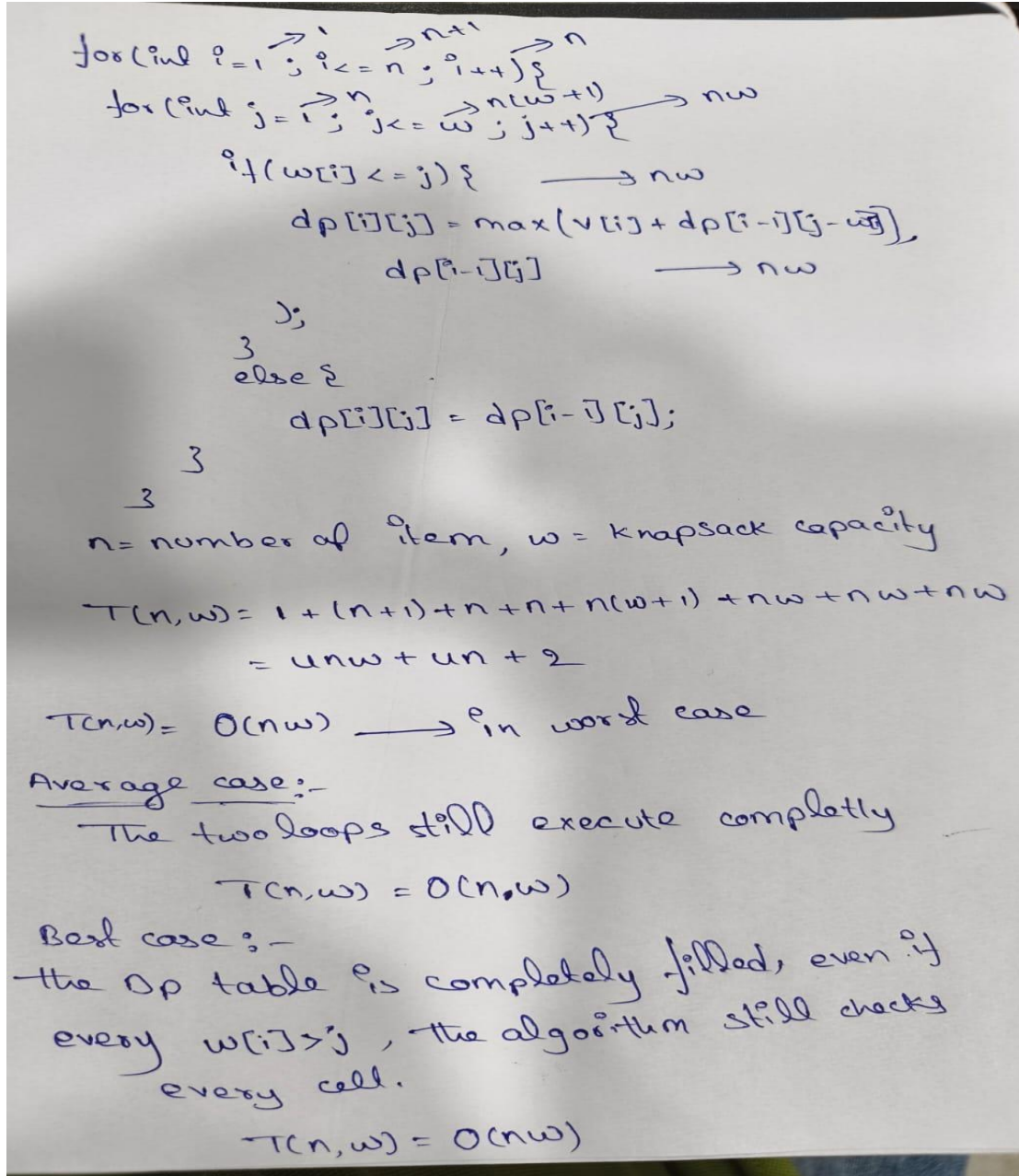
```
    }  
    }  
    }  
}  
  
void printTable() {  
    cout << "\nDP Table:" << endl;  
    cout << "    ";  
    for(int j = 0; j <= W; j++)  
        cout << j << " ";  
    cout << endl;  
    for(int i = 0; i <= n; i++) {  
        cout << "i=" << i << " ";  
        for(int j = 0; j <= W; j++)  
            cout << dp[i][j] << " ";  
        cout << endl;  
    }  
}  
  
void findItems() {  
    cout << "\nSelected Items:" << endl;  
    int j = W;  
    for(int i = n; i >= 1; i--) {  
        if(dp[i][j] != dp[i-1][j]) {  
            cout << "Item " << i  
                << " (w=" << w[i]  
                << ", v=" << v[i]  
                << ") TAKEN " << endl;  
            j = j - w[i];  
        } else {  
            cout << "Item " << i  
                << " (w=" << w[i]  
                << ", v=" << v[i]  
                << ") NOT taken " << endl;  
        }  
    }  
}
```

```
    }  
}  
void showResult() {  
    cout << "\nMaximum Value = "  
        << dp[n][W] << endl;  
}  
};  
int main() {  
    Knapsack k;  
    k.getInput();  
    k.solve();  
    k.printTable();  
    k.showResult();  
    k.findItems();  
    return 0;  
}
```

Output:

```
● PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-5\knapsnack.exe"  
Enter number of items : 4  
Enter knapsack capacity : 8  
Item 1 weight and value : 2  
1  
Item 2 weight and value : 3  
2  
Item 3 weight and value : 4  
5  
Item 4 weight and value : 5  
6  
  
DP Table:  
    0  1  2  3  4  5  6  7  8  
i=0  0  0  0  0  0  0  0  0  0  
i=1  0  0  1  1  1  1  1  1  1  
i=2  0  0  1  2  2  3  3  3  3  
i=3  0  0  1  2  5  5  6  7  7  
i=4  0  0  1  2  5  6  6  7  8  
  
Maximum Value = 8  
  
Selected Items:  
Item 4 (w=5, v=6) TAKEN  
Item 3 (w=4, v=5) NOT taken  
Item 2 (w=3, v=2) TAKEN  
Item 1 (w=2, v=1) NOT taken  
○ PS C:\MinGW\bin> █
```

Time complexity:



```

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= w; j++) {
        if (w[i] <= j) {
            dp[i][j] = max(v[i] + dp[i-1][j-w[i]],
                          dp[i-1][j]);
        }
        else {
            dp[i][j] = dp[i-1][j];
        }
    }
}

```

n = number of item, w = knapsack capacity

$$T(n, w) = 1 + (n+1) + n + n + n(w+1) + nw + nw + nw$$

$$= 4nw + 2n + 1$$

$$T(n, w) = O(nw) \rightarrow \text{in worst case}$$

Average case:-
The two loops still execute completely

$$T(n, w) = O(n, w)$$

Best case:-
the dp table is completely filled, even if every $w[i] > j$, the algorithm still checks every cell.

$$T(n, w) = O(nw)$$

Conclusion: The 0/1 Knapsack Problem was successfully implemented using Dynamic Programming. The algorithm efficiently finds the maximum value that can be obtained within the given knapsack capacity by considering whether each item should be selected or not selected. The time complexity of the algorithm is $O(nW)$.